UNIVERSITY OF CHITTAGONG
Department of Management
Bachelor of Business Administration (BBA) - 1st Semester

COURSE

BCC 110
Information Technology in Business

CHAPTER 1

Information Systems in Global Business Today

Complete Study Notes for the Semester Final Examination

Every topic explained with definition, example, scenario, facts, real-life use and a specific example. Includes chapter summary, key terms and likely exam questions with model answers.

Reference Book
Management Information Systems: Managing the Digital Firm, 17th Edition
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> ***How to use these notes:*** *Topics follow the order of the lecture slides for Chapter 1. Each topic has six parts - Definition and Explanation, Example, Short Story / Scenario, Key Facts, Real-Life Use, and a Specific Example. Slide-based facts come from the lecture slides and the reference book; the stories and scenarios are teaching illustrations to help you remember the ideas. For exam answers, always use your teacher's slide definitions as the base and add examples.*

1. The Role of Information Systems in Business Today

Definition and Explanation

An **information system (IS)** is a set of interrelated components that collect, process, store and distribute information to support decision making, coordination and control in an organization. Today almost every organization - a bank, a hospital, a university, a garments factory or a small online shop - depends on information systems for its daily work. The lecture slides explain that information systems are **transforming business** through three major developments:

- **Emerging mobile digital platform:** Smartphones, tablets and wireless networks allow employees, customers and suppliers to connect from anywhere at any time. Business is no longer tied to a desk or an office building. Mobile apps, mobile banking and mobile commerce are all products of this platform.
- **Growing business use of "big data":** Big data means extremely large and complex collections of data generated by transactions, social media, sensors and websites. It is often described by three Vs: *volume* (very large amounts), *velocity* (generated at high speed) and *variety* (text, numbers, images, video). Firms analyse it to understand customers and predict demand.
- **Growth in cloud computing:** Cloud computing means renting computing power, storage and software over the Internet instead of owning and maintaining all hardware. A firm pays for what it uses and can scale up or down quickly.

Globalization opportunities: The Internet has drastically reduced the cost of operating on a global scale. Foreign trade and outsourcing have increased. This presents **both challenges** (more competition from everywhere) **and opportunities** (access to new customers, suppliers and talent).

Example

A fashion retailer takes orders through a mobile app (mobile digital platform), studies which items customers view and buy to decide what to stock (big data), keeps its website and records on rented servers (cloud computing), and ships to customers in other countries (globalization).

Short Story / Scenario

> *Nusrat runs a small online clothing page from Chattogram. Customers order through Facebook Messenger and she receives payments through a mobile financial service. After three months she checks her sales records and notices that panjabis and sarees sell most in the two weeks before Eid, so she stocks early the next year. A customer living in Dubai finds her page and places an order. Nusrat never rented a shop, yet she used a mobile platform, her own small data, a cloud-based page and global reach - all because of information systems.*

Key Facts

- Information systems are no longer only "back-office" tools; they are central to how modern firms compete.
- The three technology shifts named in the slides are mobile digital platform, big data and cloud computing.
- The Internet lowers the cost of reaching global customers, which is why small firms can now sell internationally.
- Outsourcing (giving a business task to an outside, often foreign, firm) has grown with better communication systems.

Real-Life Use

Banks offer mobile apps for balance checks and transfers; e-commerce firms use customer data for recommendations; universities use cloud platforms for online classes; exporters use the Internet to find buyers abroad.

Specific Example of this Topic

Amazon Web Services (AWS) is a leading cloud provider that lets businesses run applications and store data without managing their own servers. **Uber** shows the mobile platform in action, and **Netflix** shows big data used to recommend shows.

2. The Digital Firm

Definition and Explanation

A **digital firm** is an organization in which nearly all significant business relationships with customers, suppliers and employees are **digitally enabled and mediated**. According to the slides, a digital firm has these features:

- Core business processes (such as taking an order, paying a supplier or hiring an employee) are accomplished through **digital networks** that span the whole organization or link several organizations.
- **Key corporate assets** - intellectual property (copyrights and patents), core competencies, financial assets and human assets - are managed digitally.
- Digital firms offer **greater flexibility** in organization and management. Work can be done *time-shifted* (at any hour) and *space-shifted* (from any place).

The slides also point out a **growing interdependence** between a firm's ability to use information technology and its ability to implement corporate strategies and achieve corporate goals.

Example

An online bank with no branches: customers open accounts, transfer money and apply for loans through an app; staff review applications on internal systems; accounting, HR and compliance records are all stored and processed digitally.

Short Story / Scenario

> *Imagine two travel agencies. The first keeps paper files and takes bookings only at its counter from 10 am to 6 pm. The second is fully online: customers book flights at midnight, payment is automatic, staff work from home and the manager checks live sales dashboards on her phone. When a sudden flight disruption happens, the second agency informs all affected travellers by message within minutes. The second agency is a digital firm; it responds faster and serves more people with the same staff.*

Key Facts

- "Digital firm" is a stage that firms move towards; very few are 100 percent digital.
- Time shifting and space shifting are the two flexibility benefits highlighted in the slides.
- Digital firms manage even intangible assets such as patents and know-how through digital systems.
- The idea is closely tied to cloud computing and the mobile digital platform.

Real-Life Use

Online retailers, digital banks, ride-sharing platforms, streaming services and many software companies operate as digital firms; traditional firms are adding digital processes to become more like them.

Specific Example of this Topic

Amazon is often cited as a digital firm: customer relationships, supplier orders, warehouse operations and even its financial assets are handled through connected digital systems.

3. Interdependence between Organizations and Information Technology

Definition and Explanation

Interdependence means that two things depend on each other. In contemporary firms there is a growing interdependence between the **business firm** (its strategic objectives and business processes) and its **information system** (hardware, software, data management and telecommunications technology). Figure 1-2 of the textbook (shown on slide 4) illustrates this two-way relationship.

This works in both directions. **First**, changes in strategy, rules and business processes increasingly require changes in hardware, software, databases and telecommunications. For example, a firm that decides to sell online needs new software and a secure payment network. **Second**, what the organization would like to do often depends on what its systems will permit it to do. If the existing system cannot handle real-time inventory, the firm cannot promise same-day delivery.

For this reason managers must understand both business and technology; strategy and information systems must be planned together, not separately.

Example

A supermarket chain decides to start home delivery (strategy change). Its old system cannot take online orders, so it must buy new software, build a website and connect it with its inventory database (change in technology).

Short Story / Scenario

> *A garments manufacturer wins a large order from a foreign buyer who demands daily online production reports. The owner wants to accept, but the factory's computers cannot generate such reports. He must either upgrade the system or lose the order. Here the system's capability limits the business's decision - a clear sign of interdependence.*

Key Facts

- Strategy influences systems, and systems influence which strategies are possible.
- Four technology areas are involved: hardware, software, data management and telecommunications.
- The slides stress that the organization's wishes "depend on what its systems will permit it to do".
- Poor alignment between business and IT is a common reason for failed projects.

Real-Life Use

Companies create IT steering committees, include IT managers in strategy meetings and plan IT budgets together with business plans so that both move in step.

Specific Example of this Topic

Banks introducing mobile banking: the strategy (reach more customers) required new software, data security systems and telecom links, while the available technology also shaped what services could be offered.

4. Information Systems

Definition and Explanation

An **information system** is "a set of interrelated components that collect, create, process, store and distribute data and information to support an organization's operations, decision-making and strategic objectives" (lecture slide 5). In simple words, it is a combination of technology, people and procedures that turns data into useful information for the organization.

Information systems **help organizations in several ways**:

- Support decision making, coordination, control and strategic planning.

- Enable data-driven problem solving and innovation.
- Facilitate collaboration and communication across global networks.
- Provide digital solutions to dynamic business and societal challenges.

Components of an IS (as listed in the slides): Hardware, Software, Data, Procedures and People. Each component is discussed in detail under the five-component framework later in this chapter.

Example

A university's student record system: it collects admission and result data (input), stores it in a database, processes it to calculate CGPA, and distributes mark sheets and reports to students and the controller of examinations.

Short Story / Scenario

> *Tania is a first-semester student who wants to know her attendance percentage. In the old days she had to visit the department office and wait while a clerk searched a register. Now the department uses an information system: attendance is entered daily, the database stores it, the system calculates the percentage and Tania sees it on her phone in seconds. The same system also helps the head of department identify students at risk of low attendance.*

Key Facts

- An IS is more than a computer; it includes people and procedures.
- The five components are Hardware, Software, Data, Procedures and People.
- An IS supports decision making at all levels of an organization.
- An IS may be manual or computerized, but in business today the term usually means computer-based information systems.

Real-Life Use

Payroll systems, hospital patient records, airline reservation systems, ATM networks, library systems and online shopping platforms are all information systems.

Specific Example of this Topic

Salesforce is a customer relationship management (CRM) system. It collects customer data, tracks interactions and supports sales, marketing and customer service - a complete example of an information system.

5. Data and Information

Definition and Explanation

Data are streams of **raw facts** representing events such as business transactions. They are the raw material from which information is generated. Data appear in the form of text, numbers, figures or any combination of these, and they must be **processed** to become useful.

Information is clusters of facts or data that have been processed and converted into a form that is **meaningful and useful** to human beings, for example in making decisions. Facts, principles, knowledge, experience and intuition are applied to convert data into information. In the slide's words: *information is data that has been transformed into a more useful form.*

The information system sits in the middle: raw data (such as a list of items sold with prices) goes in, and information (such as total yearly sales by store and product) comes out.

Example

A supermarket scanner records lines such as "Brite Dish Soap - 1.29", "BL Hill Coffee - 4.69" (data). The system adds them up and reports "Brite Dish Soap, 7,156 units sold, year-to-date sales $9,231.24 in Superstore #122" (information).

Short Story / Scenario

> *Karim, a shopkeeper, writes down every sale in a notebook: "rice 2 kg", "oil 1 litre", "sugar 1 kg". On their own these lines are only data. At month-end his son enters them into a spreadsheet and finds that oil is his fastest-selling product and rice has a low profit. Now Karim has information. He decides to order more oil and negotiate a better rice price - the data became useful only after processing.*

Key Facts

- Data = raw, unprocessed, little meaning by itself. Information = processed, meaningful, useful for decisions.
- The same data can produce different information depending on who needs it.
- Today the huge volume of data is called "big data".
- Information from one process can become data for another process.

Real-Life Use

Managers use sales reports (information) built from daily transactions (data); banks turn transaction records into statements; hospitals turn test readings into diagnoses; governments turn census data into planning information.

Specific Example of this Topic

Starbucks collects data about each sale and customer; analysis of this data gives information that helps decide where to open new stores and what to stock.

Data versus Information - Quick Comparison

Basis	Data	Information
Meaning	Raw facts and figures	Processed, meaningful facts
Usefulness	Little use alone	Directly useful for decisions
Form	Text, numbers, figures	Reports, summaries, charts
Dependency	Independent raw material	Depends on data and processing
Example	331 Brite Dish Soap 1.29	Soap sales were 7,156 units this year

6. Activities of an Information System and the Process of Converting Data into Information

Definition and Explanation

An information system performs **three basic activities** that produce the information organizations need, plus a fourth control activity:

- **Input:** captures or collects raw data from within the organization or from its external environment (for example, scanning a barcode or typing an order).
- **Processing:** converts the raw input into a meaningful form - classifying, arranging (sorting) and calculating.
- **Output:** transfers the processed information to the people or activities that will use it - reports, bills, messages, dashboards.
- **Feedback:** output that is returned to appropriate members of the organization to help them evaluate or correct the input stage.

An organization also has an **environment**. Environmental actors such as **customers, suppliers, competitors, stockholders and regulatory agencies** interact with the organization and its information systems (Figure on slide 7).

Example

Stages of converting data into information (slide 8): (1) Collection, (2) Classification, (3) Sorting, adding, merging and so on, (4) Summarizing, (5) Storing, (6) Retrieval, (7) Dissemination, (8) Action and Feedback.

Short Story / Scenario

> *A pharmacy records each medicine sold (collection) and groups sales by medicine type (classification). It sorts by date and adds totals (sorting and adding), summarizes monthly sales (summarizing), saves the file (storing), later pulls out last Ramadan's record (retrieval), and sends the report to the owner (dissemination). The owner orders more saline and tells staff to record expiry dates properly (action and feedback).*

Key Facts

- Remember the cycle: Input - Process - Output - Feedback.
- Processing includes classify, sort, summarize, store, retrieve, duplicate and disseminate.
- Feedback makes the system self-correcting; without it errors repeat.
- The process produces information; *action* on that information creates business value.

Real-Life Use

Online order systems: the order form is input, the system checks stock and calculates price (processing), the invoice and delivery slip are output, and customer complaints about wrong items return as feedback to improve data entry.

Specific Example of this Topic

Walmart's point-of-sale systems capture each purchase (input), update stock levels and sales trends (processing), produce replenishment reports (output) and managers adjust orders based on results (feedback).

7. Characteristics of Good Information

Definition and Explanation

Not all information is equally valuable. The slides list the qualities that make information good enough to support decisions:

- **Subjectivity:** The value and usefulness of information are highly subjective. What is information for one person may not be for another (for example, a change in stock price matters to an investor but not to a farmer).
- **Relevancy:** Information is good only when it is relevant and meaningful to the decision maker.
- **Timeliness:** It must be delivered at the right time, at the right place, to the right person.
- **Accuracy:** It must be free of errors at any stage of collection and processing.
- **Correct information format:** It must be presented in a format that is useful to the decision maker (for example a chart rather than a thousand rows).
- **Completeness:** It is complete if the decision maker can satisfactorily solve the problem at hand using it.
- **Accessibility:** It is useless if it is not readily accessible to decision makers in the desired format when required.

Example

A weather forecast is useful to a fisherman only if it is accurate, reaches him before he goes to sea (timely), relates to his area (relevant) and is easy to understand (format).

Short Story / Scenario

> *The sales manager of a company needs to decide on Monday morning how many units to produce. The system sends her a 40-page report on Wednesday with figures from the wrong region, in tiny tables. Although the report contains data, it fails on timeliness, relevance and format, so she ignores it and relies on guesswork. Later the IT team redesigns the report: a one-page dashboard, regional, ready every Sunday night. Her decisions improve at once.*

Key Facts

- Seven characteristics: subjectivity, relevancy, timeliness, accuracy, format, completeness, accessibility.
- Poor-quality information leads to poor decisions, even if the technology is advanced.
- Accuracy errors can occur at any stage, so controls are needed throughout.
- Information that arrives late loses its value, however accurate it is.

Real-Life Use

Companies set data quality rules, create dashboards for managers, set refresh schedules for reports and train staff on correct data entry.

Specific Example of this Topic

Stock exchanges show real-time prices because for traders timeliness and accuracy decide whether information is valuable; a price that is ten minutes old can be worthless.

8. How Information Systems Are Transforming Business

Definition and Explanation

The slides list several visible ways in which information systems are changing the way business is done:

- **Social networking tools** are used by businesses to connect employees, customers and managers. Internal platforms and public networks make communication faster and more open.
- **Internet advertising** continues to grow at more than 20 percent per year (figure given in the lecture slides), shifting marketing budgets from print and television to digital channels.
- **New laws** require businesses to store more data for longer periods, so firms must invest in storage, security and record management systems.
- **Changes in business result in changes in jobs and careers.** New roles such as data analyst and cloud engineer appear, while some routine jobs shrink.

Other slides add that wireless technology, websites and Web 2.0 technologies are used more, and that cloud computing and the mobile digital platform allow more **distributed work, decision making and collaboration**.

Example

A bank uses Facebook pages and chat to answer customer questions, runs digital advertising campaigns, stores customer records for the number of years required by regulation, and hires more IT security staff than before.

Short Story / Scenario

> *Mehedi, a marketing graduate, finds that his first job is almost entirely digital: he runs paid ads on social media, answers customers through chat and prepares weekly analytics reports. His uncle, who started in the same field twenty years ago, spent most of his time arranging newspaper adverts and printing brochures. Same field, but information systems have changed the tools, the skills and the career path.*

Key Facts

- Four changes to remember: social tools, internet advertising growth, new data-retention laws, new jobs and careers.

- Web 2.0 means interactive web tools such as social networks, blogs and wikis where users create content.
- Regulation (for example, financial record-keeping rules) makes data storage a legal duty, not only a choice.

Real-Life Use

Companies recruit through professional networking sites, hold virtual meetings, advertise on search engines and social media, and retrain staff for digital skills.

Specific Example of this Topic

Digital advertising by Facebook and Google is a clear example of how information systems moved large advertising budgets from traditional media to online platforms.

9. What Is New in Management Information Systems

Definition and Explanation

The slides summarize the latest developments in MIS in five groups:

- **IT innovations:** cloud computing, big data, Internet of Things (IoT), mobile digital platform, and artificial intelligence (AI) and machine learning.
- **New business models:** online streaming of music and video, and on-demand e-commerce services.
- **E-commerce expansion:** the slides state that e-commerce expanded to nearly $1 trillion in 2018; Netflix had more than 125 million US subscribers; online services now approach online retail in revenue; and online mobile advertising is now larger than desktop advertising.
- **Management changes:** managers use social networks and collaboration tools; business intelligence applications accelerate; virtual meetings proliferate.
- **Firms and organizations change:** they become more collaborative with less emphasis on hierarchy and structure; greater emphasis on competencies and skills; faster and more accurate decision making based on data and analysis; more willingness to interact with consumers through social media; and a better understanding of the importance of IT.

Example

A manager joins a video meeting with team members in three cities, opens a business intelligence dashboard showing live sales, and shares comments in a collaboration tool - all on one tablet.

Short Story / Scenario

> *A regional sales head used to wait for monthly printed reports and call each branch to ask about problems. Now a business intelligence dashboard updates every hour, an AI tool highlights branches that are falling behind, and a team chat group handles questions instantly. Her company also cut two management layers because information now reaches decision makers directly. The new technology changed not only her tools but also the structure of the organization.*

Key Facts

- IoT means everyday physical devices (sensors, meters, machines) connected to the Internet that send and receive data.
- Machine learning lets systems improve by learning from data rather than following fixed rules only.
- Business intelligence (BI) means tools that analyse data and present insights to managers.
- Figures such as $1 trillion e-commerce and 125 million subscribers are as given in the lecture slides (textbook edition data).

Real-Life Use

Smart factories use IoT sensors; retailers use AI for demand forecasts; managers use BI dashboards; companies stream content on demand instead of selling physical media.

Specific Example of this Topic

Netflix replaced DVD rental by mail with streaming - a new business model enabled by information systems and the Internet.

10. Five-Component Framework of Information System

Definition and Explanation

The five-component framework helps us think about information systems now and in the future. An IS is made up of five components:

- **Hardware:** physical devices such as servers, computers, IoT sensors and mobile devices.
- **Software (programs):** applications, operating systems and cloud-based platforms that process data.
- **Data:** the "lifeblood" of an IS; includes both structured data (tables, records) and unstructured information (emails, images, video).
- **Procedures:** methods, policies and rules governing data use and system operation.
- **People:** users, IT professionals and decision makers who interact with and manage the system.

In the textbook figure, hardware and programs form the *computer side*, while procedures and people form the *human side*, with data as the *bridge* between them. **Automation** occurs when a business process is moved from the human side to the computer side. The slide also notes that the degree of difficulty of change **increases** as we move toward the human side - changing a computer is easier than changing people. This is why the slide says: **"The most important component - YOU."**

Example

A hospital pharmacy system: barcode scanners and servers (hardware), pharmacy software (software), patient and medicine records (data), rules for verifying prescriptions (procedures) and pharmacists and nurses (people).

Short Story / Scenario

> *A company buys expensive new inventory software (hardware and software are ready) but the storekeepers were never trained and the old paper rules still apply. After three months the system is hardly used. The company then trains the staff and revises its procedures; usage jumps. The lesson: the human-side components decide success.*

Key Facts

- Five components: Hardware, Software, Data, Procedures, People.
- Computer side = hardware and programs; human side = procedures and people; data bridges them.
- Automation moves work from human side to computer side.
- People are the hardest component to change, and the most important.

Real-Life Use

IT managers use the framework to diagnose why a system fails: is the problem hardware, software, data quality, unclear procedures or untrained people?

Specific Example of this Topic

Bank ATM network: ATMs (hardware), banking software, account data, withdrawal rules and limits (procedures), customers and bank staff (people).

11. Information System versus Information Technology

Definition and Explanation

The slides distinguish clearly between these two terms.

Information System (IS) is a coordinated set of components and resources designed to collect, store, process and distribute information. It typically involves computer hardware and software, databases and communication systems that help organizations manage operations, interact with customers and make informed decisions. *Example in the slides: a Customer Relationship Management (CRM) system, such as Salesforce, which helps manage customer data, track interactions and improve customer service.*

Information Technology (IT) refers to the use of computers, networks, storage and other physical devices, infrastructure and processes to create, process, store, secure and exchange all forms of electronic data. IT covers the hardware and software needed to manage and operate information systems. *Example in the slides: cloud computing, such as Amazon Web Services, which lets businesses run applications and store data without managing physical servers.*

The key difference (slide 13): **an information system incorporates the technology, people and processes involved with information; information technology is the design and implementation of information or data within the information system.** In short, IT is a part (a tool) of an IS; an IS is the bigger whole.

Example

A university uses servers, network equipment and database software (IT). The complete student management system, including staff who enter results and the rules for publishing them, is the information system.

Short Story / Scenario

> *Two friends argue. Sabbir says, "We bought new laptops and a router, so we have an information system." Tania replies, "That is only technology. Without the software, the data, the rules for using it and the staff who run it, it does not yet help the business decide anything." Tania is right: IT is the toolbox; IS is the working system built from the tools, people and procedures.*

Key Facts

- IT = tools (hardware, software, networks); IS = tools + people + processes + data.
- All information systems use IT, but IT alone is not an information system.
- The slides use CRM (Salesforce) for IS and cloud computing (AWS) for IT.

Real-Life Use

IT departments manage infrastructure and security; business units use information systems such as CRM, ERP and accounting systems to run their processes.

Specific Example of this Topic

Salesforce (IS) runs on cloud infrastructure from providers such as **AWS (IT)** - the two layers work together.

Comparison Table

Basis	Information System (IS)	Information Technology (IT)
Scope	Broader - technology, people, procedures, data	Narrower - hardware, software, networks
Purpose	Support operations, decisions, strategy	Provide tools and infrastructure
Focus	Information and its use in the organization	Technology that creates, stores and moves data
Slide example	CRM such as Salesforce	Cloud computing such as AWS

12. Why Study Information Systems?

Definition and Explanation

The slides give several reasons why a business student must study information systems:

- To become an **educated end-user** who can use systems effectively and critically.
- To **contribute to the design** of new systems in your future job.
- To **increase organizational success**.
- To **advance your career**.
- To meet the challenges of a **global market** where management decisions become more complex.
- To learn how data and information are used by the organization - **information systems literacy**.

The slide also quotes a warning from *Data Communications*: a lack of understanding by upper management about how to oversee, budget and understand the real expense and value of the enterprise information infrastructure costs companies millions of dollars in cost overruns and failed attempts to improve productivity and launch new products. In other words, managers who do not understand information systems make costly mistakes.

Example

A finance manager who understands information systems asks the right questions before approving a new accounting software - cost, risks, training needs, expected benefits - and avoids paying for features the firm will never use.

Short Story / Scenario

> *Two managers are asked to approve a software project. Manager A signs without questions because "IT knows best". The project runs over budget and nobody uses the system. Manager B asks for a clear business case, a training plan and a pilot test. Her project finishes on budget and staff adopt it. Studying information systems gives managers the confidence to act like Manager B.*

Key Facts

- Information systems literacy is part of management competence today.
- Cost overruns and failed projects often come from management's lack of understanding, not only from technical faults.
- Business graduates are expected to work with IT specialists in teams.

Real-Life Use

Business graduates work as analysts, project coordinators, marketers using analytics, auditors checking systems, and entrepreneurs building digital businesses.

Specific Example of this Topic

Banking graduates who understand core banking systems can suggest improvements to customer service and reduce errors, making them more valuable to employers.

13. Why Organizations Need Information Systems

Definition and Explanation

The slides state that few, if any, medium- or large-sized companies can survive without computers and information systems. The world has become a small place and competition is no longer limited by boundaries. The reasons organizations need information systems are:

- **Meeting global challenges:** produce high-quality goods and services that can compete in world markets.
- **Capturing opportunities in the marketplace:** successful companies identify and take advantage of opportunities through strategic information systems (SIS).

- **Supporting corporate strategy:** using the three basic strategies - cost strategy, product differentiation and market niche strategy.
- **Linking departments whose functions are different:** accounting, finance, marketing, manufacturing and human resources are connected through shared systems.
- **Enhancing worker productivity:** computers and information systems have a significant impact on a company's bottom line.
- **Increasing the quality of goods and services:** for example through Total Quality Management (TQM) promoted by top and middle managers.

Example

An ERP system links the sales department, the factory and the accounts department: when sales enters an order, production sees it immediately and accounts prepares the invoice automatically.

Short Story / Scenario

> *In a mid-sized furniture company, the sales team promised delivery in two weeks, but the factory did not know about the order until a week later, and accounts did not know the price agreed. Customers complained and costs rose. After the company installed an integrated system, all three departments saw the same order data at the same time. Delivery promises became realistic and complaints fell. This shows why firms need information systems to link departments.*

Key Facts

- Seven needs: global challenges, market opportunities, strategy, linking departments, productivity, quality (and survival in competition).
- Three basic strategies: cost leadership, product differentiation, market niche.
- Strategic information systems (SIS) are systems that help a firm gain competitive advantage.
- TQM = Total Quality Management, a firm-wide effort for continuous quality improvement.

Real-Life Use

Manufacturers use ERP, retailers use point-of-sale and inventory systems, banks use core banking systems, airlines use reservation systems.

Specific Example of this Topic

Walmart links suppliers, warehouses and stores through its supply chain systems and so keeps prices low (cost strategy).

14. Globalization Challenges and Opportunities: A Flattened World

Definition and Explanation

The phrase "flattened world" means that the Internet and global communications have levelled the playing field, so that firms and individuals anywhere can compete and cooperate across the globe. The slides explain that the Internet and global communications have **greatly changed how and where business is done**. The main points are:

- Drastic reduction of the cost of operating and transacting on a global scale.
- Competition for jobs, markets, resources and ideas.
- Growing interdependence of global economies.
- New understandings of skills, markets and opportunities are required.
- There are challenges, but also opportunities.
- Over half of the revenue of S&P; 500 US firms is generated offshore (slide fact).
- **Information systems enable the globalization of commerce.**

Example

A software firm in Dhaka serves clients in Europe through video calls, shared online project tools and online payment - without opening any office abroad.

Short Story / Scenario

> *Farhana is a freelance graphic designer in Chattogram. Ten years ago her clients were limited to local advertising agencies. Today she works for a start-up in Canada, a restaurant chain in Malaysia and an author in Australia, receiving projects and payment online. But she also competes with talented designers from many countries and must keep improving her skills and pricing. Globalization has brought her both new opportunities and tougher competition.*

Key Facts

- "Flattened world" is a term popularised by Thomas Friedman in his book *The World Is Flat*.
- Challenges: competition, cultural and legal differences, need for new skills. Opportunities: bigger markets, cheaper resources, global talent.
- Information systems (email, video, cloud platforms, e-commerce) are the infrastructure of globalization.

Real-Life Use

Outsourcing call centres, global supply chains in garments and electronics, remote teams, cross-border e-commerce and international payments.

Specific Example of this Topic

Bangladesh's ready-made garments (RMG) exporters use online systems to receive buyer orders, share designs and track shipments across countries.

15. How Information Systems Transform Business Today

Definition and Explanation

The slides summarize the transformation under two heads.

1. Creating globalization opportunities. The Internet has reduced the cost of operating on a global scale for both customers and firms. Using Internet service firms such as Google and eBay, businesses can replicate their business and services in multiple countries without a large additional investment.

2. Helping the emerging digital firm. Business relationships in a digital firm are digitally enabled and mediated. Its **core business processes** - sets of logically related tasks and behaviors, such as developing a new product, generating and fulfilling an order, creating a marketing plan or hiring an employee - are accomplished through digital networks. Key corporate assets (intellectual property, core competencies, financial and human assets) are managed through digital means.

Other slides add: the increase in wireless technology and websites, increased business use of Web 2.0 technologies, and cloud computing and the mobile digital platform allowing more distributed work, decision making and collaboration.

Example

A small seller lists products on an international marketplace such as eBay and starts receiving orders from several countries without opening a single foreign office.

Short Story / Scenario

> *A young entrepreneur in Dhaka designs leather wallets. He lists them on a global online marketplace; the platform handles search, payment and customer reviews. Within six months he has customers in four countries. His business process of "generating and fulfilling an order" is done through digital networks: the order arrives online, payment is confirmed electronically, the courier is booked through an app. He did not need to invest in shops abroad - the information systems of the platform did the global work for him.*

Key Facts

- A business process is a set of logically related tasks and behaviors for accomplishing work.
- Examples of business processes: developing a product, fulfilling an order, creating a marketing plan, hiring an employee.
- Platforms such as Google and eBay let firms replicate services in many countries with limited extra investment.

Real-Life Use

E-commerce marketplaces, digital advertising, online recruitment, digital payment gateways and cloud-based accounting for small firms.

Specific Example of this Topic

Google and eBay (named in the slide) offer worldwide reach to any business that uses their platforms.

16. Six Strategic Business Objectives of Information Systems - Overview

Definition and Explanation

Business firms invest heavily in information systems to achieve **six strategic business objectives**. This explains *why firms use IS*. The slides emphasise the growing interdependence between the ability to use information technology and the ability to implement corporate strategies and achieve corporate goals. The six objectives are:

- **1. Operational excellence** - improvement of efficiency to attain higher profitability.
- **2. New products, services and business models** - enabled by technology.
- **3. Customer and supplier intimacy** - serving customers well raises revenues and profits.
- **4. Improved decision making** - more accurate data leads to better decisions.
- **5. Competitive advantage** - doing better than rivals.
- **6. Survival** - keeping up with industry standards, regulations and market demands.

A useful memory trick is to build a sentence from the first letters: **O**perational, **N**ew products, **C**ustomer, **I**mproved decision, **C**ompetitive advantage, **S**urvival. The following six topics explain each objective with the examples from the lecture slides.

Example

A retail company installs an integrated system: it saves cost (operational excellence), launches online sales (new business model), uses customer data to personalize offers (customer intimacy), analyses sales for better decisions, gains an edge over rivals and meets tax and reporting rules (survival).

Short Story / Scenario

> *The owner of a growing electronics shop wonders whether to spend on a new computer system. His consultant lists what the system can do: cut stock losses, allow online selling, remember each customer's preferences, show weekly profit by product, help him beat nearby competitors and keep tax records in the format the government now demands. Each point matches one of the six objectives, so he approves the investment.*

Key Facts

- The six objectives appear in almost every MIS exam; learn their names in order.
- The first four are about better business performance; the last two are about market position and long-term continuation.
- The textbook examples are Toyota and FedEx (operational excellence), Netflix and Apple (new models), Amazon (intimacy), Starbucks and Zara (decisions), Walmart and Nike (advantage), Kodak

(survival).

Real-Life Use

Every sector - banking, retail, health, education, manufacturing - justifies IT investment by linking it with one or more of these objectives.

Specific Example of this Topic

Amazon demonstrates almost all six objectives together, which is why the slides use it as a case study.

17. Objective 1: Operational Excellence

Definition and Explanation

Businesses continuously seek to improve the **efficiency** of their operations in order to achieve higher profitability. Information systems and technology are among the most important tools for achieving greater efficiency and productivity - they streamline processes, reduce errors and cut costs.

Example

Toyota uses its Just-In-Time (JIT) inventory system to reduce waste and improve manufacturing efficiency, which boosts profitability. **FedEx** uses advanced logistics and tracking systems to optimize its delivery network, improving efficiency and reducing costs - leading to faster deliveries and higher profitability (slide examples).

Short Story / Scenario

> *A bakery owner used to count flour and sugar by eye and often ran out or over-ordered. After he started a simple stock-tracking sheet linked to daily sales, he ordered exactly what was needed. Waste fell, cakes were always available and profit rose. Efficiency came from information, not from more labour.*

Key Facts

- Just-In-Time means receiving materials only when needed for production, so storage cost and waste are low.
- Operational excellence focuses on doing the same things *better, faster and cheaper*.
- Typical measures: cost per unit, delivery time, error rate, productivity.

Real-Life Use

Barcode and RFID tracking in warehouses, automated billing, online ticketing, route optimization for deliveries.

Specific Example of this Topic

FedEx tracks every package through scanners and systems, letting customers see location in real time and the company reroute efficiently.

18. Objective 2: New Products, Services and Business Models

Definition and Explanation

Information technology enables firms to **innovate**: to create new products and services and to develop entirely new **business models**. A business model describes how a company produces, delivers and sells a product or service to create wealth.

Example

Netflix transformed the entertainment industry by using streaming technology, creating a new business model and disrupting traditional media consumption. **Apple** used information systems to introduce the App Store, a new business model that allows developers to sell apps directly to consumers; this transformed the mobile industry and created a new revenue stream (slide examples).

Short Story / Scenario

> *Before ride-hailing apps, a person needing a taxi stood on the roadside or phoned a company. Then a start-up connected drivers and riders through a smartphone app that handled location, fare and payment. Nobody needed to own a fleet of cars; the platform itself was the business. Technology made a model that did not exist before.*

Key Facts

- New business models often disrupt existing industries (music, video, taxis, hotels).
- Platform models connect two groups, such as developers and users or drivers and riders.
- Netflix started in 1997 as a DVD-by-mail service and launched streaming in 2007.
- The App Store opened in 2008.

Real-Life Use

Digital banking, online education, app stores, subscription streaming, mobile wallets and on-demand delivery.

Specific Example of this Topic

Apple App Store and **Netflix streaming** as given in the slides.

19. Objective 3: Customer and Supplier Intimacy

Definition and Explanation

When a business knows its customers well and serves them well, customers generally return and spend more, raising revenues and profits. In the same way, close relationships with suppliers help a firm get better prices and reliable supply. Information systems let businesses understand and respond to customer needs and strengthen supplier relationships. **Intimacy** here means close, data-supported knowledge of the other party.

Example

Amazon uses personalized recommendation systems based on customers' browsing and purchase data, leading to increased sales and customer loyalty. It also strengthens supplier relationships through efficient order management systems (slide example).

Short Story / Scenario

> *A neighbourhood bookshop owner in Chattogram remembers each regular customer's favourite authors and calls when a new book arrives. Customers love it - but she cannot do this for thousands of people. An online bookstore uses its system to do the same for millions: "Customers who bought this also bought..." The system gives personal attention at large scale.*

Key Facts

- CRM (Customer Relationship Management) systems are the main tool for customer intimacy.
- Supplier intimacy is supported by supply chain management systems and shared data with suppliers.
- Customer loyalty is cheaper to keep than new customers are to win.

Real-Life Use

Loyalty cards, personalized offers, customer service chat histories, supplier portals and automatic reordering.

Specific Example of this Topic

Amazon's recommendation engine and its supplier order management systems.

20. Objective 4: Improved Decision Making

Definition and Explanation

Without accurate, real-time data, managers rely on forecasts, best guesses and luck, which leads to poor decisions, wrong stock levels or wrong pricing. Information systems provide access to accurate, real-time data so that managers can make better-informed decisions, which improves outcomes and profits.

Example

Starbucks uses sales and customer data to optimize store locations and inventory, improving decision making and profitability. **Zara** uses real-time sales data collected from its stores to quickly adjust inventory and design new clothing lines, enabling data-driven decisions about what products to stock and produce (slide examples).

Short Story / Scenario

> *A clothing store manager sees from the system that red kurtas sell out within two days while blue ones stay on the shelf. Instead of ordering equal numbers, she orders more red and discounts blue. A month later, her sales are higher and her unsold stock is lower. The decision was not luck - it was based on data.*

Key Facts

- Better data does not guarantee better decisions, but poor data nearly always harms them.
- Decision support types: transaction processing, management information, decision support and executive systems (covered later in the textbook).
- Zara is known for a very fast cycle between design and store; its sales data drives production.

Real-Life Use

Dashboards, sales forecasting, credit scoring in banks, site selection for new branches.

Specific Example of this Topic

Zara and **Starbucks** as above.

21. Objective 5: Competitive Advantage

Definition and Explanation

A firm achieves **competitive advantage** when it does something better, faster or cheaper than its rivals, or offers something rivals cannot easily copy. Information systems help firms gain a competitive edge through **differentiation, cost leadership or innovation**. Often the firm that responds to customers faster, delivers cheaper or personalizes better wins customers from competitors.

Example

Walmart uses its advanced supply chain management system to maintain lower prices and faster delivery, outperforming competitors. **Nike** uses data analytics and digital platforms (like Nike Training Club and Nike Run Club) to engage customers, gather insights and offer personalized experiences, which differentiates it from competitors in the athletic wear industry (slide examples).

Short Story / Scenario

> *Two tea stalls stand side by side. One serves the same tea but with an app where customers order ahead and skip the queue, earning points for every cup. Slowly, office workers shift to the second stall. The tea is the same; the information system created the advantage.*

Key Facts

- Advantage can come from low cost (Walmart), differentiation (Nike) or innovation.

- Advantage from technology can be temporary if rivals copy it, so firms must keep innovating.
- This objective links directly to the three basic strategies: cost, differentiation, niche.

Real-Life Use

Faster delivery promises, loyalty apps, personalized service, online ordering, supply chain efficiency.

Specific Example of this Topic

Walmart (cost leadership) and **Nike** (differentiation).

22. Objective 6: Survival

Definition and Explanation

Sometimes firms invest in information systems not to gain an edge but simply to **survive**. Information systems are essential for keeping up with industry standards, complying with government regulations and responding to market demands. If competitors offer online banking, a bank must also offer it or lose customers; if the law requires electronic records, firms must comply.

Example

Kodak failed to adapt to digital photography and the shift to online photo storage, which ultimately led to its downfall, while companies like **Canon** and **Nikon** adapted their business models, incorporating digital technology to survive and thrive (slide example).

Short Story / Scenario

> *When mobile banking became common, a small bank decided it was too costly. Younger customers began moving their accounts to banks with apps. Within two years the bank lost a large part of its customer base. It finally built an app, but much damage was done. Some technology investments are not luxuries; they are the price of staying in business.*

Key Facts

- Survival reasons: industry standards, regulations, customer expectations.
- Kodak engineers invented a digital camera in 1975, but the company did not fully shift its business model, and it filed for bankruptcy protection in 2012.
- The ATM is a well-known example: banks adopted it because rivals did.

Real-Life Use

Regulatory reporting systems, e-tax filing, digital payment acceptance, cybersecurity systems.

Specific Example of this Topic

Kodak versus Canon and Nikon.

23. Solved Case Study: Amazon and the Six Objectives

Definition and Explanation

Case (slide 23): Amazon continuously invests in advanced information systems to manage its vast supply chain, improve customer recommendations and enhance operational efficiency. These systems help Amazon achieve faster deliveries, personalized services and data-driven decisions.

Question: Explain how Amazon's use of information systems helps the company achieve operational excellence, customer intimacy, improved decision making, competitive advantage and overall corporate success.

Model Answer:

- **Operational excellence:** Amazon's inventory management and warehouse systems track stock levels in real time and keep products available; automated order processing and optimized delivery routing cut handling time and cost. Efficiency raises profit.

- **Customer intimacy:** Its recommendation system uses browsing and purchase data to personalize what each customer sees, increasing sales and loyalty. Its order management systems also strengthen relationships with suppliers.
- **Improved decision making:** Data analytics systems analyse customer behaviour and demand predictions so managers decide what to stock, where, and how much - reducing overstock and stock-outs.
- **Competitive advantage:** Faster deliveries and personalized services are difficult for rivals to match; low costs and speed differentiate Amazon from competitors.
- **Overall corporate success:** Together, these systems raise revenue, cut costs and build customer trust and loyalty, strengthening long-term strategic position and growth. Information systems therefore are not an add-on but the engine of Amazon's business model.

Example

See the model answer above; it applies the objectives to one firm.

Short Story / Scenario

> *A customer in Chattogram ordering a book from an online store sees recommendations that suit her taste (intimacy), finds the item in stock (operational excellence), receives it quickly (advantage) and the company's managers use thousands of such orders to plan next month's stock (decisions). The customer returns again and again.*

Key Facts

- Amazon started in 1994 as an online bookstore and expanded into many product lines and cloud services.
- The slides use Amazon as an example in objectives 3, the business perspective, managerial assets and technical approaches.

Real-Life Use

Any firm can apply this answer structure: name the objective, link it to a system, show the business result.

Specific Example of this Topic

Exam tip: Answer case questions in four steps - define the objective, name the Amazon system, explain how it works, state the business benefit.

24. Computer Literacy versus Information Literacy

Definition and Explanation

Computer literacy is the ability to efficiently use computers and digital technologies to perform tasks, solve problems and communicate. It involves understanding hardware, software, networks and cloud systems, plus basic knowledge of cybersecurity, AI tools and data management. Computers act as extensions of human intelligence, improving productivity and creativity. It focuses on using digital tools confidently and safely in personal and professional contexts.

Information literacy is the ability to **locate, evaluate and use information effectively** through digital and information systems to support decision making and gain a competitive advantage. It builds on computer literacy and adds **critical thinking and data literacy**. It requires understanding how information systems support organizational goals and involves skills in data analysis, digital communication and ethical use of information. It enables individuals to **turn data into insights** for innovation and strategic success.

Example

A student who can open spreadsheet software, type formulas and create a chart is computer literate. If she also knows which data to collect, checks that the source is reliable, interprets the chart correctly and uses the result ethically to propose a business decision, she is information literate.

Short Story / Scenario

> *Two interns receive the same sales file. Intern A formats the sheet beautifully and prints it. Intern B questions the numbers: "Why do March sales look 40 percent lower? Is data missing from one branch?" She finds the missing data, corrects it and discovers the real trend. Both know the software; only Intern B shows information literacy.*

Key Facts

- Computer literacy focuses mainly on knowledge of IT (tools).
- Information literacy focuses on using information for decisions, with critical thinking.
- The slides state that MIS tries to achieve this broader information literacy.

Real-Life Use

Employers look for graduates who can not only use software but also analyse data, judge sources and communicate insights.

Specific Example of this Topic

A manager reading a dashboard and deciding to cut a slow product line is applying information literacy.

Comparison Table

Basis	Computer Literacy	Information Literacy
Meaning	Ability to use computers and digital tools	Ability to locate, evaluate and use information
Focus	Technology and tools	Information, analysis and decisions
Skills	Hardware, software, networks, cybersecurity basics	Critical thinking, data analysis, ethical use
Goal	Productivity and safe use of tools	Insight, innovation, competitive advantage
Relation	Foundation	Builds on computer literacy

25. Dimensions of Information Systems

Definition and Explanation

To fully understand information systems, one must understand the broader **organization**, **management** and **information technology**. These three are the **core perspectives (dimensions)** needed to understand how an IS functions within an organization. Understanding these dimensions is **information system literacy** (broader, behavioral as well as technical), while computer literacy focuses primarily on knowledge of IT. MIS tries to achieve the broader information literacy.

MIS deals with **behavioral issues** as well as **technical issues** surrounding the development, use and impact of information systems used by managers and employees in the firm. In the textbook figure, Organizations, Management and Technology surround Information Systems at the center.

- **Organizations:** people, structure, business processes, politics, culture.
- **Management:** decisions, strategy, resource allocation, creativity.
- **Technology:** hardware, software, data management, networking.

Example

When a bank adopts a new loan system, success depends on the technology (software and servers), the organization (loan officers, approval rules, culture) and management (strategy, budget, leadership support).

Short Story / Scenario

> *A hospital buys modern patient-record software. The technology works perfectly, but doctors resist typing notes (organization), and the director never explained why the system matters or allocated training time (management). Six months later records are still half on paper. The failure was not technical; two dimensions were ignored.*

Key Facts

- Three dimensions: Organization, Management, Technology.
- A system is more likely to fail if only the technology dimension is considered.
- This three-dimension model is the standard way to analyse a case in an exam.

Real-Life Use

Project teams use the three dimensions to plan system implementation: check technology fit, organizational readiness and management support.

Specific Example of this Topic

Delta Bank case (solved below) uses exactly these three dimensions.

26. Organizational Dimension of Information Systems

Definition and Explanation

Information systems are an essential part of modern organizations. In many industries, such as credit reporting, finance and e-commerce, businesses cannot operate without them. The key elements of an organization are **people, structure, business processes, politics and culture**, and these influence how information systems are used and managed.

Hierarchy of levels (Figure 1-6). Most organizations have a hierarchical structure with defined roles and responsibilities:

- **Senior management** makes long-range strategic decisions about products and services and ensures the financial performance of the firm.
- **Middle management** carries out the programs and plans of senior management; **operational management** is responsible for monitoring the daily activities of the business.
- **Knowledge workers** (engineers, scientists, architects) design products or services and create new knowledge for the firm; **data workers** (secretaries, clerks) assist with scheduling and communications at all levels.
- **Production or service workers** actually produce the product and deliver the service.

Digital technologies today link all levels of the organization, improving communication, efficiency and decision making.

Major business functions (Table 1-2): Sales and marketing (selling the organization's products and services); Manufacturing and production (producing and delivering products and services); Finance and accounting (managing financial assets and records); Human resources (attracting, developing and maintaining the labour force and employee records).

Business processes are logically related tasks and behaviors for accomplishing work, such as developing a new product, fulfilling an order or hiring a new employee. Some are formal (written rules, such as writing an invoice) and some are informal work practices (such as returning phone calls promptly). Information systems automate many business processes - for instance, how a customer receives credit or is billed is often determined by an information system.

Culture is the fundamental set of assumptions, values and ways of doing things accepted by most members. **Politics**: different levels and specialties have different interests and points of view that often conflict over how resources and rewards should be distributed; information systems come from this mix of differing perspectives, conflicts and compromises.

Example

In a university, the registrar's office (a function) uses an online admission system (IS) to process applications; the process (receive form - verify - rank - notify) is automated; the culture of strict deadlines affects how staff use the system.

Short Story / Scenario

> *A factory introduces a system that tracks each worker's output hourly. Senior managers love the reports. Production supervisors fear being blamed, so they enter inflated numbers. The system then gives false information. The new system had collided with the organization's culture and politics - a lesson that organizational factors strongly shape IS success.*

Key Facts

- Five key elements: people, structure, business processes, politics, culture.
- Levels: senior management, middle management, operational management; plus knowledge workers, data workers and production/service workers.
- Four major functions: sales and marketing, manufacturing and production, finance and accounting, human resources.
- Business processes can be formal or informal.

Real-Life Use

Implementation teams map existing business processes before automating them, involve all levels and manage resistance.

Specific Example of this Topic

Credit approval in a bank: the billing and credit rules are written into the system; the bank's risk-averse culture shapes how strict they are.

27. Management Dimension of Information Systems

Definition and Explanation

Management's job is to make sense out of the many situations faced by organizations, make decisions and formulate action plans to solve organizational problems. According to the slides, **managers perceive business challenges in the environment; they set the organizational strategy for responding to those challenges; and they allocate the human and financial resources to coordinate the work and achieve success.**

In addition, managers must act **creatively** and do more than what already exists: *creating new products and services* and occasionally *re-creating the organization*. Information systems give managers the information and tools to monitor, decide and innovate.

Example

A retail chain's manager sees online competition rising (perceives challenge), decides to launch its own online store (sets strategy) and assigns budget and a project team (allocates resources).

Short Story / Scenario

> *The CEO of a family-owned pharmacy chain notices customers increasingly ordering medicines online. She studies the market, decides to launch a home-delivery app, approves a budget and appoints a manager to lead the project. After launch she keeps adjusting based on reports. In just a few steps she performed the full management dimension: perceive, set strategy, allocate resources, and create something new.*

Key Facts

- Management functions in the slides: perceive challenges, set strategy, allocate resources.

- Managers must also be creative: new products/services and re-creating the organization.
- Lack of management support is a leading reason for IS failure.

Real-Life Use

Executives use dashboards for strategy; middle managers use reports for planning and control; operational managers use transaction data.

Specific Example of this Topic

Amazon's leaders supported data-driven decision making and experimentation (see managerial assets).

28. Technology Dimension of Information Systems

Definition and Explanation

Information technology is one of many tools managers use to cope with change. The slides define the technology dimension through the following components:

- **Hardware:** physical components used for inputs, processing and output.
- **Software:** preprogrammed instructions that control hardware.
- **Data management technology:** software that organizes data (databases).
- **Networking and telecommunications technology:** hardware plus software that links computers together.
- **Internet:** the "global network of networks". An **intranet** is an internal corporate network based on Internet technology. An **extranet** is a private intranet extended to authorized users outside the organization (for example suppliers).
- **World Wide Web (WWW):** a service for storing, retrieving, formatting and displaying information in a page.
- **IT infrastructure:** the platform that a firm builds its specific information systems on.

Example

A company intranet lets employees view HR policies; its extranet lets suppliers check purchase orders; both use Internet technology but with different access rules.

Short Story / Scenario

> *An exporter allows its fabric supplier to log into a restricted web page to see production schedules and upcoming orders. Employees inside the company use a similar internal portal that is invisible to outsiders. The first is an extranet, the second an intranet. Both rely on the same technology - the difference is who can enter.*

Key Facts

- The Internet and the World Wide Web are not the same: the Web is a service that runs on the Internet.
- Intranet = inside only; extranet = inside plus authorized outsiders.
- IT infrastructure is the foundation on which applications are built.

Real-Life Use

Organizations build networks, databases, cloud services and security systems as their IT infrastructure.

Specific Example of this Topic

Delta Bank's AI threat detection, multi-factor authentication and cloud monitoring are technology-dimension components.

29. Solved Case Study: Delta Bank and the Three Dimensions

Definition and Explanation

Case (slide 32): Delta Bank launched a new cybersecurity management system using AI-driven threat detection, multi-factor authentication and cloud-based monitoring to safeguard customer information. After implementation: (1) many employees find the new login and verification process complicated, leading to frequent lockouts; (2) some departments continue to use outdated data-sharing methods, creating security gaps; (3) the IT team is overwhelmed managing alerts and system updates due to limited automation and training.

Question: Analyse the case using the technical, organizational and management dimensions; identify key issues for each and suggest practical solutions.

Example

See the table of analysis below.

Short Story / Scenario

> *Imagine you are the new IS consultant hired to review Delta Bank's system. You interview staff, watch the help desk queue, and review security logs. You find that the technology is strong, but staff are confused, old habits remain and the IT team is overloaded. You present your three-dimension analysis to the board.*

Key Facts

- The question asks for issues and solutions for **each** dimension - always use headings.
- Technology alone cannot ensure security; people and management decisions matter equally.

Real-Life Use

This format (issue - cause - solution) works for any IS case analysis.

Specific Example of this Topic

Use the table below as your model answer.

Dimension	Key Issues	Practical Solutions
Technical	Complicated multi-factor login causes lockouts; alert volume is high; limited automation of alert handling and updates.	Simplify authentication (single sign-on, biometric or app-based approval); self-service password reset; automate alert triage and patching with AI/automation tools; set clear alert priorities.
Organizational	Some departments keep using outdated data-sharing methods; resistance to change; no culture of security discipline; weak procedures.	Standard data-sharing procedures through approved secure channels; phase out old methods with deadlines; department security champions; regular staff training; user-friendly guides.
Management	Insufficient training and resources for the IT team; weak change management; lack of monitoring of adoption.	Allocate budget for staff and IT team training; announce clear policy with leadership support; assign a project owner; track KPIs such as lockouts, incidents and adoption rate; gather employee feedback and improve.

Conclusion: Delta Bank's technology is advanced, but effectiveness depends on usable design, organizational discipline and management support. A balanced, sociotechnical approach will improve adoption and security.

30. Business Perspective on Information Systems

Definition and Explanation

Managers and business firms invest in information technology and systems because they provide **real economic value** to the business. The decision to build or maintain an information system assumes that the returns on this investment will be superior to other investments in buildings, machines or other assets. From a business perspective, **an information system is an important instrument for creating value for the firm.** Information systems enable the firm to **increase its revenue or decrease its costs** by providing information that helps managers make better decisions or that improves the execution of business processes.

Investments in information technology result in superior returns: productivity increases, revenue increases, superior long-term strategic positioning, data-driven decision making and competitive advantage.

An information system is not just technology - it is a solution that aligns technology with business strategies. It helps organizations operate efficiently, make informed decisions, manage resources and achieve competitive advantage by collecting, processing and analysing data.

Key elements of the business perspective (slide 34):

- **Organizational goals:** IS supports objectives such as increasing revenue, reducing costs or improving customer satisfaction.
- **Business processes:** IS improves workflows by automating tasks, reducing errors and ensuring consistency.
- **Decision making:** IS provides data and analytics for informed strategic decisions.
- **Competitive advantage:** IS can differentiate a business through unique products/services, lower operational costs or better customer experiences.

Example

Amazon (slide 35): Goal alignment - fast, reliable customer service through efficient supply chain management and personalized recommendations. Enhanced business processes - inventory management tracks stock in real time. Improved decision making - data analytics analyse customer behaviour and adjust inventory based on demand predictions. Competitive advantage - delivery systems with optimized routing ensure faster delivery than competitors.

Short Story / Scenario

> *A factory owner compares two investments of equal cost: a new machine that raises output by 10 percent, or an integrated system that cuts waste, speeds orders and improves decisions. After calculating, he sees that the information system gives a higher long-term return and also supports growth. From a business perspective, investment is justified by value creation, not by the attraction of technology.*

Key Facts

- Core idea: IS creates value by raising revenue or lowering cost.
- The value of IS is judged against other investment options.
- A system must align with business strategy; technology for its own sake fails.

Real-Life Use

Managers prepare business cases with expected benefits, costs, payback period and risks before approving IS projects.

Specific Example of this Topic

Amazon - goal alignment, processes, decisions and advantage as listed above.

31. The Business Information Value Chain

Definition and Explanation

Every business has an **information value chain** (Figure 1-7) in which raw information is systematically acquired and then transformed through various stages that add value to that information. The value of an information system to a business, and the decision to invest in any new system, is in large part determined by the extent to which the system leads to better management decisions, more efficient business processes and higher firm profitability.

In the figure, the information-processing activities are: **data collection and storage, transformation into business systems, dissemination**. These feed **business processes** (supply chain management, enterprise management, customer management, knowledge management) and **management activities** (planning, coordinating, controlling, modeling and decision making), which finally lead to **firm profitability and strategic position**. The business perspective calls attention to the organizational and managerial nature of information systems.

Example

Netflix and the value chain (slides 38-40):

- **Data collection:** what users watch, when and how long, pauses/rewinds/skips, ratings and reviews, device usage data.
- **Data processing:** categorizing viewing habits by genre, time of day and demographics; clustering users by preference (for example "comedy lovers", "true crime fans"); filtering out noisy data.
- **Information generation:** predicting what a user will enjoy, identifying trends such as the rising popularity of a genre, deciding which original content to produce or license.
- **Business processes:** streamlining production of Netflix Originals; preloading likely-to-be-watched content.
- **Business value creation:** customer satisfaction and retention, revenue growth, competitive advantage over rivals such as Hulu and Disney+, and cost efficiency by avoiding costly content failures.

Short Story / Scenario

> *Think of the value chain like making flour into bread. Wheat (raw data) is collected, ground (processed), baked (information generation), sold in shops (business processes) and the bakery earns profit (value). Each stage adds value; skipping any stage reduces the final worth.*

Key Facts

- The chain shows how data becomes profit step by step.
- Value is created only when information is used to improve decisions or processes.
- Two groups of activities: information-processing activities and management activities.

Real-Life Use

Retailers analyse loyalty-card data, banks analyse transactions for fraud and credit, and streaming firms analyse viewing data for content decisions.

Specific Example of this Topic

Netflix as described above.

32. Complementary Assets: Organizational Capital and the Right Business Model

Definition and Explanation

Information technology investments alone cannot make organizations and managers more effective unless they are accompanied by supportive values, structures and behavior patterns in the organization and other **complementary assets**. Some firms fail to adopt the right business model that suits the new technology, or seek to preserve an old business model that is doomed by new technology. *For instance, recording label companies refused to change their old business model based on physical music stores; as a result, online legal music sales are dominated not by record companies but by a technology company, Apple.*

Complementary assets (Teece, 1988) are the resources, capabilities or organizational elements that must be present for a primary investment, such as IT, to deliver maximum value. Without them the primary investment may not yield significant benefits. *For instance, to realize value from automobiles requires substantial complementary investments in highways, roads, gasoline stations, repair facilities and a legal regulatory structure to set standards and control drivers.*

Example

A company installs a very expensive ERP system but keeps its old, slow approval hierarchy and has no trained users. The system gives little value because supporting assets are missing.

Short Story / Scenario

> *A city introduces electric buses (the primary investment), but there are few charging stations, no trained mechanics and no rules for charging tariffs. The buses spend most days parked. Only after charging stations, training and policy support are added do the buses deliver their promised value. The same is true of information systems.*

Key Facts

- Three types of complementary assets: organizational, managerial and social.
- The record-label example shows the danger of keeping an old business model.
- The automobile example is the standard textbook analogy.

Real-Life Use

Before approving an IT project, successful firms also invest in training, process redesign, incentives and culture.

Specific Example of this Topic

Apple versus record labels for business model failure; **automobiles** for complementary investments.

Types of Complementary Assets (Table 1-3 and slides 43-45)

Type	Meaning and elements	Slide example
1. Organizational assets	Internal structures that let technology work effectively: supportive culture valuing efficiency, efficient business processes, appropriate business model, decentralized authority, distributed decision-making rights, strong IS development team.	Toyota Production System: success due not only to IT but also efficient processes and a culture of continuous improvement (Kaizen), where employees at all levels identify inefficiencies.
2. Managerial assets	Leadership capabilities and management practices that guide technology use: strong senior management support, incentives for innovation, teamwork and collaboration, training programs for decision making, flexible knowledge-based management culture.	Amazon's leaders support data-driven decision making, promote innovation and flexibility (experiments with AI and robotics), and provide continuous training.
3. Social assets	Exist outside the organization in society or industry: Internet and telecom infrastructure, IT-enriched education, standards and regulations, fair stable markets, support from technology and service firms.	bKash (Bangladesh): depends on mobile network coverage, government regulations supporting digital finance, and public computer literacy and trust in mobile technology.

33. Contemporary Approaches to Information Systems - Overview

Definition and Explanation

The study of information systems is a **multidisciplinary field**; no single theory or perspective dominates. In general, the field can be divided into **technical approaches** and **behavioral approaches** (Figure 1-9), and a combined **sociotechnical** view. The disciplines are:

- **Technical approach:** computer science, management science, operations research.
- **Behavioral approach:** sociology, psychology, economics.
- **Sociotechnical approach:** combines both.

Information systems are **sociotechnical systems**: though they are composed of machines, devices and "hard" physical technology, they require substantial social, organizational and intellectual investments to make them work properly. Contemporary approaches recognize that technology alone is not enough; by combining technical expertise with behavioral insight, organizations can build systems that are both efficient and user-friendly.

Example

Amazon's predictive inventory management (technical) paired with its emphasis on customer satisfaction (behavioral) shows the synergy of the two approaches.

Short Story / Scenario

> *Two teams design a school app. Team A, all engineers, builds a fast and secure system but never talks to teachers; teachers find it confusing. Team B includes engineers, teachers and a psychologist; they test with real users and revise the screens. Team B's app is adopted by every school. Combining disciplines produced a better system.*

Key Facts

- Technical approaches emphasise models and technology; behavioral approaches emphasise people and organizations.
- Neither alone is enough - this is the central message of the sociotechnical view.
- MIS itself is described as an interdisciplinary field.

Real-Life Use

Modern system projects use mixed teams: software engineers, business analysts, designers and organizational change specialists.

Specific Example of this Topic

Amazon (technical + behavioral synergy).

34. Technical Approaches to Information Systems

Definition and Explanation

The **technical approach** focuses on the formal models and capabilities of information systems, emphasizing technology and efficiency. It uses mathematically based models to study information systems and the physical technology and formal capabilities of these systems. The contributing disciplines are **computer science, management science and operations research**.

- **(a) Management science approach:** developing mathematical and statistical models to optimize decision making and operations.
- **(b) Computer science approach:** the study of computing, software and hardware to design efficient information systems.
- **(c) Operations research approach:** applying mathematical techniques to solve organizational problems such as logistics, scheduling and resource allocation.

Example

- **Management science - Amazon:** uses machine learning algorithms to optimize inventory; predictive analytics ensures warehouses stock the right products based on customer demand, minimizing overstock and understock.
- **Computer science - Google Search Engine:** its information retrieval system is built on advanced computer science principles such as algorithms and data structures, ensuring fast and accurate search results.
- **Operations research - FedEx:** optimizes delivery routes by analysing traffic, weather and distance, ensuring faster and more cost-efficient deliveries.

Short Story / Scenario

> *A delivery company has 200 parcels and 20 riders. Instead of assigning riders by guess, its software uses an operations research model to find the shortest total route. Fuel use drops, parcels arrive faster and riders finish earlier. A mathematical model solved a practical business problem.*

Key Facts

- Technical approach: models, algorithms, efficiency.
- Three disciplines: computer science, management science, operations research.
- Limitation: it does not fully explain issues such as user resistance or organizational politics.

Real-Life Use

Route planning, inventory optimization, search algorithms, scheduling of staff and machines, forecasting models.

Specific Example of this Topic

Amazon, Google and FedEx as listed in the example block.

35. Behavioral Approaches to Information Systems

Definition and Explanation

The **behavioral approach** focuses on the human and organizational aspects of information systems, ensuring that systems are user-friendly, ethical and aligned with business goals. An important part of the IS field is concerned with behavioral issues that arise in the development and long-term maintenance of systems - strategic business integration, design, implementation, utilization and management - which cannot be explored usefully with the models of the technical approach. The contributing disciplines are:

- **(a) Sociology approach:** studying the impact of information systems on individuals, groups and organizations.
- **(b) Psychology approach:** examining how individuals interact with systems to design user-friendly interfaces and reduce cognitive load.
- **(c) Economics approach:** analysing the cost-benefit aspects of information systems to ensure value creation.

Example

- **Sociology - Microsoft Teams:** designed to foster collaboration; sociological studies of team dynamics help refine video calls, shared documents and chat threads.
- **Psychology - Apple's iPhone interface:** intuitive design such as gesture-based navigation and minimalist layouts rooted in understanding user psychology.
- **Economics - Netflix:** uses economic models to assess the cost of content delivery against the value of customer satisfaction; investments in streaming infrastructure are based on economic justification.

Short Story / Scenario

> *A company introduces a new attendance system. Employees complain that the screen is confusing (psychology: reduce cognitive load), team leaders feel the system damages trust (sociology: group impact), and the finance manager asks whether the benefit justifies the cost (economics). A behavioral analysis addresses all three concerns, whereas pure technical analysis would miss them.*

Key Facts

- Behavioral approach: people, groups, organizations, user-friendliness, cost-benefit.
- Three disciplines: sociology, psychology, economics.
- It explains why technically good systems can still be rejected by users.

Real-Life Use

User testing, change management, training programs, usability design and cost-benefit analysis before approval.

Specific Example of this Topic

Microsoft Teams, iPhone interface and Netflix as listed above.

36. Sociotechnical Systems Approach

Definition and Explanation

A **sociotechnical system** is an approach to understanding the interaction between people (social factors) and technology (technical factors) in an organization. The sociotechnical view holds that **optimal organizational performance is achieved by jointly optimizing both the social and technical systems used in production.** Both social and technical elements must be considered together; in Figure 1-10, technology and organization mutually adjust to each other through alternatives until a satisfactory fit is obtained.

Adopting a sociotechnical perspective helps avoid a purely technological approach. For instance, the fact that information technology is rapidly declining in cost and growing in power does not necessarily or easily

translate into productivity enhancement or bottom-line profits. Likewise, the fact that a firm has recently introduced new business procedures does not necessarily mean employees will be more productive in the absence of investments in new information systems to enable those processes.

Example

Tesla Autopilot (slide 51): Technical component - sophisticated AI algorithms, sensors and cameras. Behavioral component - ensuring drivers are aware of system limitations and providing interfaces that encourage safe usage (for example hands-on-wheel alerts).

Uber (slide 53): Technical system - app, GPS, payment system and back-end servers match riders with drivers, calculate fares and process payments in real time. Social system - drivers and passengers interact through the app; customer support helps with problems. Integration - success depends on both; drivers who cannot navigate the app or disputes between drivers and passengers hurt overall performance.

Short Story / Scenario

> *A hospital buys a high-end scanner (technical). The radiology staff are trained, the booking process is redesigned and doctors agree on new reporting rules (social). Patients receive faster diagnoses. If the hospital had only bought the machine, waiting times would not improve.*

Key Facts

- Sociotechnical = joint optimization of technology and organization.
- Cheaper and more powerful IT does not automatically improve productivity.
- Technology and organization must adjust to each other until a satisfactory fit is reached (Figure 1-10).

Real-Life Use

Software rollout plans include system testing, user training, process changes and feedback loops - not only installation.

Specific Example of this Topic

Uber and Tesla Autopilot as described.

37. Potential Risks of Information Systems

Definition and Explanation

Information systems bring great benefits but also carry risks that managers must understand. The slides list the following:

- **Deskilling of workers:** while IS can increase operational efficiency and improve profits, it can sometimes be the root cause of workforce reduction or the loss of traditional skills.
- **Information overload:** IS generate reports easily and quickly, but this can lead to an excess of information.
- **Employee mistrust:** companies use systems to monitor employee activities, which may cause mistrust and dislike.
- **Increased competitive pressure:** as IS develop, large companies with greater investment may push small companies out of competition.
- **Misinvestment in IS:** investing in inappropriate technology, or poor application of good technology, results in poor returns and loss to the firm.
- **Challenges in developing IS:** a difficult task, because of time and budget constraints.
- **Security breaches:** the risk of protecting assets from theft and hacking.

Example

An office installs software that records each employee's keystrokes and screen time. Staff feel watched, morale falls and some resign (employee mistrust). The manager also receives so many daily reports that she cannot read them (information overload).

Short Story / Scenario

> *A small local bookshop competes well until a large online retailer with advanced systems offers lower prices and faster delivery. The bookshop cannot afford similar technology and loses customers. Meanwhile, another company buys an expensive system that nobody needs and wastes money. Both stories show risks of IS: competitive pressure and misinvestment.*

Key Facts

- Seven risks in the slides - memorise the names.
- Many risks are managed by planning, training, policy and security controls.
- Security breaches can damage reputation and lead to legal penalties.

Real-Life Use

Firms conduct risk assessments, set monitoring policies with employee consent, train staff, limit report volume to key indicators and invest in cybersecurity.

Specific Example of this Topic

Delta Bank case: security gaps from outdated data-sharing methods illustrate the security breach risk.

38. Management Information Systems (MIS)

Definition and Explanation

MIS as a field of study is an interdisciplinary field influenced by computer science, political science, psychology, operations research, linguistics, sociology and organizational theory and behavior. **MIS as an information system** is a broad class of systems that provides information to facilitate organizational decision making.

MIS is the development and use of information systems that help businesses achieve their goals and objectives. It has three key elements: **components of an information system; development and use of information systems; achieving business goals and objectives.** MIS deals with behavioral issues as well as technical issues. In the slides: **"A management information system (MIS) is a system that provides information needed to manage organizations effectively. Management information systems involve three primary resources: technology, information and people."**

An MIS collects, processes, stores and disseminates information to help in managerial decision making. It integrates data from different sources, processes it and provides insights to support decision making at various levels of the organization.

Example

Walmart's inventory and sales system (slide 56):

- **Data collection:** sales, inventory levels and customer preferences through sensors, barcode scanners and loyalty programs.
- **Data processing:** analyses sales trends, inventory data and customer behaviour to identify patterns.
- **Information storage:** a centralized database accessible to managers and executives.
- **Information dissemination:** reports on sales, stock levels and market trends.
- **Decision support:** managers use the data for replenishment, sales forecasting and product placement.

Short Story / Scenario

> *A district sales manager of a mobile phone company receives a weekly MIS report: which models sold best, which shops ran out of stock and which regions fell behind target. Based on it she shifts stock between shops and plans promotions. Without the MIS she would be guessing.*

Key Facts

- Three primary resources of MIS: technology, information, people.

- MIS is both a field of study and a type of system.
- Output of MIS: routine reports, summaries and exception reports for managers.

Real-Life Use

Sales reports, inventory reports, attendance and payroll summaries, budget variance reports and customer-service dashboards.

Specific Example of this Topic

Walmart - streamlined operations, reduced costs and data-driven decisions.

39. MIS Trends in Business

Definition and Explanation

The slides list the main current trends in MIS, with examples:

Example

- **AI integration:** boosts decision making, automates tasks and personalizes customer experiences. *Amazon uses AI to optimize inventory and recommend products, increasing sales and reducing costs.*
- **Cloud computing:** reduces infrastructure cost, enables scalability and improves collaboration. *Salesforce integrates CRM functions to enhance customer relationships and productivity.*
- **Big data analytics:** provides insights into customer behaviour and operational performance. *Walmart uses big data to optimize inventory and improve supply chain efficiency.*
- **Blockchain:** enhances data security and transparency, reducing fraud and increasing trust. *IBM Food Trust tracks food products via blockchain for transparency and safety.*
- **Real-time data processing:** enables quick responses to market changes and operational disruptions. *Uber uses real-time data to optimize ride matching and routes.*
- **Mobile MIS:** provides on-the-go access to data for faster decision making. *Microsoft Power BI enables executives to access business insights from mobile devices.*

Short Story / Scenario

> *A logistics manager, while travelling, opens her phone, sees a live map of delayed shipments (real-time, mobile), receives an AI suggestion to reroute trucks (AI), and checks the tamper-proof record of one shipment (blockchain). Four trends in one working day.*

Key Facts

- These innovations enhance business efficiency, improve customer experience and enable quicker, data-driven decision making (slide conclusion).
- Blockchain is a shared, tamper-resistant record of transactions.
- Many trends rely on each other: cloud supports big data, which supports AI.

Real-Life Use

Banks use AI for fraud detection, hospitals use cloud records, supply chains use blockchain tracking and managers use mobile dashboards.

Specific Example of this Topic

The six named examples above - Amazon, Salesforce, Walmart, IBM Food Trust, Uber and Microsoft Power BI.

Chapter Summary

Information systems have become essential to modern business. Three major technology shifts - the mobile digital platform, big data and cloud computing - together with globalization are transforming how firms compete. The Internet lowers the cost of global operations, increasing foreign trade and outsourcing, which creates both challenges and opportunities (a "flattened world").

A **digital firm** manages its significant business relationships, core business processes and key corporate assets digitally, and gains flexibility through time shifting and space shifting. There is growing **interdependence** between a firm's strategy and its information systems.

An **information system** is a set of interrelated components - **hardware, software, data, procedures and people** - that collect, process, store and distribute information. **Data** are raw facts; **information** is processed, meaningful data. The system performs **input, processing, output and feedback**, and good information is accurate, timely, relevant, complete, accessible, correctly formatted and valued subjectively. **IT** is the technology; an **IS** includes technology, people and processes.

Firms invest in IS to achieve **six strategic business objectives**: operational excellence; new products, services and business models; customer and supplier intimacy; improved decision making; competitive advantage; and survival.

To understand IS fully, one must study three **dimensions**: **organization** (people, structure, business processes, politics, culture), **management** (perceiving challenges, setting strategy, allocating resources, creativity) and **technology** (hardware, software, data management, networking, internet, intranet, extranet, WWW, IT infrastructure). Information system *literacy* goes beyond computer literacy.

From a **business perspective**, an IS is an instrument for creating value by raising revenue or lowering cost, as seen in the **business information value chain**. IT investments pay off only when supported by **complementary assets** - organizational, managerial and social.

IS is studied through **technical approaches** (computer science, management science, operations research), **behavioral approaches** (sociology, psychology, economics) and the **sociotechnical approach** that jointly optimizes technology and organization. Risks include deskilling, information overload, employee mistrust, competitive pressure, misinvestment, development difficulty and security breaches.

MIS provides the information needed to manage organizations effectively using three resources - technology, information and people - and current trends include AI, cloud computing, big data, blockchain, real-time processing and mobile MIS.

Key Terms and Definitions

Term	Meaning
Information system (IS)	Interrelated components that collect, process, store and distribute information to support decisions and control.
Information technology (IT)	Computers, networks, storage and related infrastructure used to create, process, store and exchange data.
Data	Raw facts representing events such as business transactions.
Information	Data processed into a meaningful and useful form for decisions.
Input / Processing / Output / Feedback	The four activities of an IS: capture, convert, deliver, and return results to evaluate or correct input.
Digital firm	Organization where relationships, core processes and key assets are digitally enabled and managed.
Mobile digital platform	Smartphones, tablets and wireless networks enabling anywhere, anytime business.

Term	Meaning
Big data	Extremely large, fast and varied data sets analysed for insight (volume, velocity, variety).
Cloud computing	Renting computing power, storage and software over the Internet.
Globalization / flattened world	Worldwide integration of business enabled by the Internet, lowering costs and increasing competition.
Outsourcing	Giving a business task to an outside (often foreign) provider.
Five components of IS	Hardware, software, data, procedures, people.
Business process	Logically related tasks and behaviors for accomplishing work.
Operational excellence	Improving efficiency to achieve higher profitability.
Customer and supplier intimacy	Close, data-supported relationships that raise revenue and profit.
Competitive advantage	Doing better than rivals through differentiation, cost leadership or innovation.
Strategic information system (SIS)	System that helps a firm capture market opportunities and gain advantage.
Computer literacy	Ability to use computers and digital tools effectively and safely.
Information literacy	Ability to locate, evaluate and use information to support decisions.
Three dimensions of IS	Organizations, management and technology.
Knowledge worker / data worker	Creates new knowledge for the firm (engineer) / supports scheduling and communication (clerk).
Organizational culture	Fundamental assumptions, values and ways of doing things accepted by an organization's members.
Organizational politics	Conflict and compromise over resources and rewards among groups with different interests.
Intranet / Extranet	Internal Internet-based network / private intranet extended to authorized outsiders.
IT infrastructure	Platform on which a firm builds its specific information systems.
Business information value chain	Series of stages that transform raw data into information that adds value to the business.
Complementary assets	Resources and organizational elements needed for IT investment to deliver full value.
Sociotechnical system	View that technology and social systems must be optimized together.
Deskilling	Reduction in the skill level of work due to automation.
Management information system (MIS)	System providing information needed to manage organizations effectively, using technology, information and people.
Blockchain	Shared, tamper-resistant record of transactions that increases trust and transparency.

Likely Exam Questions with Model Answers

Q1. Define information system. Explain its components.

An **information system** is a set of interrelated components that collect, create, process, store and distribute data and information to support an organization's operations, decision making and strategic objectives. Its five components are: (1) **Hardware** - physical devices such as computers, servers, sensors; (2) **Software** - programs, operating systems and cloud platforms; (3) **Data** - structured and unstructured facts, the lifeblood of the system; (4) **Procedures** - rules and policies governing use; (5) **People** - users, IT staff and decision makers, the most important component. Example: a university result system.

Q2. Distinguish between data and information with examples.

Data are raw facts and figures representing events, such as "331 Brite Dish Soap 1.29". They have little meaning alone. **Information** is processed, meaningful data useful for decisions, such as "Brite Dish Soap sold 7,156 units this year, sales $9,231.24". Data are the raw material; information is the output; processing converts one into the other. Information depends on the user: the same data may be valuable to one person and useless to another.

Q3. Describe the activities of an information system.

An IS has input (capturing raw data from inside or outside the organization), processing (classifying, arranging, calculating), output (delivering processed information to users as reports or messages) and feedback (output returned to appropriate people to evaluate or correct input). The environment - customers, suppliers, competitors, stockholders and regulators - interacts with the organization and its IS. Example: a supermarket billing system.

Q4. What are the characteristics of good information?

Good information is **relevant** to the decision maker, **timely** (right time, place and person), **accurate** (error-free), in the **correct format**, **complete** (enough to solve the problem) and **accessible** when needed. Its value is also **subjective**: information for one person may not be for another. Poor-quality information leads to poor decisions.

Q5. Why do organizations need information systems?

Organizations need IS to meet global challenges, capture market opportunities through strategic information systems, support corporate strategy (cost, differentiation, niche), link different departments such as accounting, finance, marketing, manufacturing and HR, enhance worker productivity and improve the quality of goods and services (for example through TQM). Few medium or large firms can survive without IS in a competitive, borderless world.

Q6. Explain the six strategic business objectives of information systems with examples.

(1) **Operational excellence** - efficiency and profitability; Toyota JIT, FedEx logistics. (2) **New products, services and business models** - Netflix streaming, Apple App Store. (3) **Customer and supplier intimacy** - Amazon recommendations and supplier order systems. (4) **Improved decision making** - Starbucks store location data, Zara real-time sales data. (5) **Competitive advantage** - Walmart supply chain, Nike digital platforms. (6) **Survival** - meeting standards and regulations; Kodak failed to adapt while Canon and Nikon survived.

Q7. Differentiate between computer literacy and information literacy.

Computer literacy is the ability to use computers and digital technologies efficiently, focusing on hardware, software, networks, basic cybersecurity and tools. Information literacy is the ability to locate, evaluate and use information effectively to support decisions and gain advantage; it builds on computer literacy and adds critical thinking, data analysis and ethical use. Computer literacy is knowledge of IT; information literacy (IS literacy) is the broader understanding that MIS aims at.

Q8. Discuss the three dimensions of information systems.

Organizations - people, structure (hierarchy of senior, middle and operational managers, knowledge, data and production workers), business processes, politics and culture; major functions are sales and marketing, manufacturing, finance and accounting, HR. **Management** - perceives challenges, sets strategy, allocates human and financial resources and acts creatively. **Technology** - hardware, software, data management, networking and telecommunications, Internet, intranet, extranet, WWW and IT infrastructure. IS success requires all three.

Q9. What are complementary assets? Explain their types with examples.

Complementary assets are resources and organizational elements required to derive value from a primary investment such as IT; without them the investment may not yield significant benefits (like roads and fuel stations for cars). Types: **Organizational** (supportive culture, efficient processes, decentralized authority, strong IS team) - Toyota Production System and Kaizen; **Managerial** (senior management support, incentives, teamwork, training) - Amazon's data-driven leadership; **Social** (Internet infrastructure, regulations, education, stable markets) - bKash depends on mobile networks, regulation and public trust.

Q10. Explain the business information value chain with an example.

It is a series of stages in which raw data is acquired and transformed to add value: data collection and storage, transformation into business systems, dissemination; these support business processes (supply chain, customer, enterprise and knowledge management) and management activities (planning, coordinating, controlling, decision making), leading to firm profitability and strategic position. Netflix collects viewing data, processes it into clusters, generates insights on what to produce, improves operations and gains customer satisfaction, revenue, advantage and cost efficiency.

Q11. Explain the technical and behavioral approaches to information systems.

The **technical approach** uses formal mathematical models and technology; its disciplines are computer science (Google search), management science (Amazon inventory models) and operations research (FedEx routes). The **behavioral approach** focuses on people and organizations; its disciplines are sociology (Microsoft Teams), psychology (iPhone interface design) and economics (Netflix cost-benefit). The **sociotechnical approach** combines them to jointly optimize technology and the social system (Uber, Tesla Autopilot).

Q12. Discuss potential risks of information systems.

Risks include deskilling of workers and workforce reduction, information overload, employee mistrust due to monitoring, increased competitive pressure on small firms, misinvestment in inappropriate technology, challenges in developing systems due to time and budget constraints, and security breaches such as theft and hacking. Management should plan carefully, train staff, set clear policies and invest in security.

Q13. What is MIS? Explain with an example and mention current trends.

MIS is a system that provides information needed to manage organizations effectively, involving technology, information and people; it is also an interdisciplinary field of study. Walmart's inventory and sales system collects data through scanners and loyalty programs, processes sales trends, stores data centrally, generates reports and supports replenishment decisions. Current trends: AI integration, cloud computing, big data analytics, blockchain, real-time data processing and mobile MIS.

Q14. Distinguish between information system and information technology.

IT refers to computers, networks, storage and infrastructure used to create, process, store and exchange data - for example, cloud computing via AWS. An IS is a coordinated set of components (technology, people, procedures, data) that collects, stores, processes and distributes information to support decisions - for example, a CRM such as Salesforce. IS incorporates technology, people and processes; IT is the design and implementation of technology within the IS.

Q15. Case: Amazon / Delta Bank.

See the solved case studies in this chapter: the Amazon case (operational excellence, customer intimacy, decision making, competitive advantage, corporate success) and the Delta Bank case (technical, organizational and management issues with solutions). Practise writing them in your own words with headings.

> ***Final exam tips:*** *(1) Start each answer with the slide definition. (2) Add at least one real-world example from the slides. (3) Use headings and short points for list-type questions. (4) For cases, answer by dimension or objective, and end with a conclusion. (5) Revise the six objectives, five components, three dimensions and seven characteristics of good information - they are the most repeated points.*